

Pest-ConFormer: A hybrid CNN-Transformer architecture for large-scale multi-class crop pest recognition

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ABSTRACT

Crop pests are acknowledged as the major factors in reducing the yield and quality of agricultural production worldwide. It is an urgent necessity to recognize crop pests accurately to protect the crop in the early stage to reduce the loss for the agricultural economy. Due to the ecological characteristics of the crop pests and the complex background in fields, the crop pests show high inter-class similarity and significant intra-class variation in external morphology appearance, which makes current recognition methods suffer from low classification accuracy and poor generalization ability in complex natural environment recognition tasks. To tackle this problem, a hybrid convolutional neural network and transformer-based model, namely Pest-ConFormer, featured with multi-scale weakly supervised feature selection mechanisms is proposed, which has shown excellent multi-scale discriminative feature extraction in fine-grained visual classification (FGVC) tasks. This method employs a hybrid convolution-transformer encoder architecture pre-training in a self-supervised masked autoencoder manner as a backbone to learn pests' highly discriminative features across various scales. Next, a dual-path feature aggregation structure with a top-down FPN-like feature pathway and a bottom-up PANet-like feature pathway based on attention mechanisms is designed to learn high-level global context information and low-level local detailed feature representation. Thirdly, a fine-grained classification module using weakly supervised learning is introduced to select the discriminative feature points in different pyramidal levels. Then, these feature points are fed into a graph convolutional network to accomplish classification. Several experiments are conducted on the large-scale multi-class IP102 benchmark dataset, and the proposed method achieves an accuracy of 77.81 % regarding crop pest recognition. The experimental results indicate that our approach outperforms other state-of-the-art methods by nearly 2 percent points, demonstrating that the proposed hybrid architecture with dual-path feature aggregation and fine-grained classification modules can be more effective in the crop pest recognition field than CNN-based methods and can be deployed in the practical natural environment. The source code will be available at <https://github.com/mwfang/pestconformer>.

1. Introduction

Agriculture is the basic industry of a country, and how to improve crop yields has become one of the most significant worldwide issues. Crop pests and diseases can cause considerable losses to agriculture crop production. Accurate knowledge of the categories of crop pests and diseases as well as their severity is crucial for preventing and controlling

crop pests and diseases (Zhou et al., 2021). Under most situations, when the pest is correctly identified, crop damage has already occurred in the field. Therefore, identifying pest severity and taking protective measures are essential to increase crop yields and quality. Traditional crop pest prevention and control rely on agriculture experts' rich knowledge and experience. Due to the limited number of experts, various crop pests, and irregular distribution of pest damage spots, pest control and prevention

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suffer from problems of low efficiency, slow progress, and high cost, which seriously limits modern agriculture's development. Modern agriculture urgently needs efficient, accurate, and low-cost crop pest identification technology, which has strong practical significance for quickly identifying and classifying large-scale crop pests.

With the rapid development of computer vision technology, machine learning and deep learning technologies have been widely applied to crop pest recognition in recent years. Traditional machine learning method extracts visual features, such as texture, color, and shape, from visible light, hyperspectral, and near-infrared spectra images of crop pests. Then, some traditional classification algorithms such as Support Vector Machine, k-nearest Neighbor and Artificial Neural Networks are used to recognize crop pests, which improves efficiency compared to conventional manual recognition methods. Larios et al., (2010) used Haar random forests to extract features, and an SVM classifier with a specialized non-linear kernel was applied to identify aquatic stonefly larvae; this approach got state-of-the-art accuracy with higher efficiency. Faithpraise et al., (2013) first used color space transformation structure in the images, then adopted k-means clustering with a correspondence filter to identify the plant pests. Kaya and Kayci (2014) obtained the texture features using a level co-occurrence matrix and identified the butterfly species using ANN. Their classification accuracy reached 92.85 %. The above methods mainly rely on traditional machine learning methods, and their recognition performance excessively depends on the design of visual features and classifiers. However, there are various crop pests, and the same pest species may have different shapes, colors, sizes, and backgrounds in different life cycles. Therefore, traditional machine learning methods have some disadvantages, such as complex feature parameter extraction and poor generalization ability, leading to poor recognition performance in natural environments.

Compared to conventional machine learning, the deep learning method can automatically extract features layer by layer through a deep convolutional neural network. Its recognition speed and accuracy are better than the machine learning method, which provides a feasible pathway for preventing and controlling crop pests. Cheng et al., (2017) designed an identification model based on deep residual learning to identify 10 agricultural pests. The accuracy of recognition under complex farmland backgrounds was significantly improved compared to the classic CNN models. Alves et al., (2020) designed a deep residual network named ResNet34*, which was obtained using a series of linear and batch normalization operations to replace the last linear layer in the ResNet34 model. Combined with transfer learning and data augmentation strategy, this model achieved the highest accuracy in the cotton pest classification field. Chen et al., (2021) created a MobileNet-V2-based network, combined with a spatial and channel attention mechanism with a classification activation map, to identify plant pests with cluttered backgrounds, training by a progressive learning strategy. The proposed procedure achieved high accuracy and efficiency. Xing and Lee (2022) designed a novel CNN backbone named Decoupling-and-Attention network, training with the TLDP method. This model achieved the highest classification accuracy in the comprehensive image dataset of crop pests and diseases. In addition to these works for the whole crop pest image classification, some researchers focused on localizing and identifying each pest. For instance, Liu et al., (2021) proposed a new network to localize and detect each pest instance in the field. They integrated a globally activated feature pyramid network and a locally activated region proposal network into a two-stage CNN to monitor pests. Their model can accurately and effectively detect 16 types of pests in the field. Recently, transformer-based models have gradually been applied in computer vision, benefiting from the advantage of the encoder-decoder architecture of transformers in feature extraction (Vaswani et al., 2017). Vision Transformer (ViT) (Dosovitskiy et al., 2020) applied transformer architecture in image classification tasks for the first time. When pre-trained on large-scale datasets and then fine-tuned on small-scale data sets for the classification task, the ViT model achieved better results and outperformed the best CNN

classification networks.

In summary, the deep learning method for crop pest identification can be divided into coarse-grained and fine-grained classification. The former denotes the cross-species semantic-level classification with many dissimilarities, while the latter denotes the subcategory classifications with massive similarity. Because of the unique ecological characteristics, crop pest classification in natural environments can be considered a challenging FGVC task. On the one hand, many environmental factors exist, such as uneven lighting, dense occlusion, and background interference in crop pests' images, which makes the same category of crop pests appear differently in various images. Meanwhile, the same category of crop pest shows different appearances in different growth stages, leading to the fine-grained characteristic of intra-class variance. On the other hand, the different subcategories of crop pests may have similarities in morphology and habitats, resulting in the fine-grained recognition problem of inter-class similarity. Existing coarse-grained classification and transfer learning methods cannot effectively recognize large-scale multi-class crop pests with fine-grained characteristics in agricultural scenarios, and these methods may have experienced technical challenges such as low recognition accuracy and poor robustness. As a result, it's necessary to enhance the model ability of discriminative feature representation and accurate region localization to learn tiny local differences among the different categories of crop pests. Common FGVC methods can be divided into strongly and weakly supervised learning (Wei et al., 2022). The former requires much manual annotation information, such as bounding box, coordinate, and label information, limiting its practical application. Weakly supervised learning methods only need images and label information. It can detect the class-discriminative regions via a classification task without additional annotation information and then utilize those discriminative regions to distinguish objects. Inspired by this, we adopt this strategy in our new approach.

We have developed a crop pest FGVC model by combining a convolutional neural network (CNN) with a transformer architecture to address the unique and intricate challenges of large-scale multi-class agricultural pest FGVC problems. Our approach incorporates the concepts of locality, inductive bias, and shift-invariance from CNN into the transformer model. This allows the transformer to maintain its capacity to model long-range dependencies using its attention mechanism. Hence, the primary contributions of this study can be briefly described in multiple aspects:

- (1) A pre-trained hybrid convolution-transformer encoder architecture is deployed as a backbone to extract highly discriminative features at various levels and scales.
- (2) A dual-path feature aggregation structure with a top-down feature pathway and a bottom-up attention feature aggregation pathway is introduced to learn high-level global context information and low-level local detailed feature representation.
- (3) A fine-grained classification module is proposed to select the most discriminative features in different pyramidal levels. These selected features are then fused into several super nodes as global discriminative features through a Graph Convolutional Network to accomplish classification.

2. Related works

2.1. CNN-Transformer hybrid methods

As a mainstream model in the field of computer vision, the CNN model has the characteristics of inductive bias, such as translation invariance and local sensitivity, which can capture the local detail information well. However, due to the limited receptive field, the CNN model lacks the capability to model long-range dependencies between spatial domains. Furthermore, the network weights of the model are fixed after the training, which cannot dynamically adapt to the changes

of the input. Inspired by the success of the transformer model on language tasks, several recent studies have applied transformers to computer vision tasks. Compared to CNN, the transformer can model global dependencies of images by computing the attention weights between each pair of features to obtain an updated feature mapping, where each position contains information about any other feature in the same image. Therefore, the transformer can dynamically adjust its receptive field according to the image size, which makes it easier to obtain global information.

ViT model utilized the classical transformer encoder structure to accomplish the image classification task, which was the beginning of the visual transformer, breaking the framework limitations of traditional CNN in computer vision and providing a new paradigm for visual feature learning. However, there still exist several shortcomings in ViT model. Such as depending on large-scale datasets, lack of capability to learn local features, and expensive computation costs. In order to enhance the locality of the transformer, the most straightforward approach was to combine the advantages of both convolutions and transformers (Maaz et al., 2022). These hybrid models, known as the CNN-Transformer model, introduced the inductive biases inherent in CNNs into the transformers while preserving their capabilities of global information modeling. Various kinds of hybrid architectures were proposed in this domain to incorporate CNNs with transformers in different ways (Khan et al., 2023). These methods typically included, (a) using several convolutional layers to extract local features at early stages, and then sequentially feeding these feature maps into transformers to capture long-range global contextual information (Wu et al., 2021; Guo et al., 2022), (b) integrating CNN backbone with self-attention module (Srinivas et al., 2021), (c) utilizing a parallel path method to combine CNN and transformer, with an additional module to fuse their predictions (Wu et al., 2023). CvT (Wu et al., 2021) introduced two independent convolution operations into token embeddings and feature projection of the transformer architecture, respectively. These convolution operations in CvT were able to learn various local spatial patterns in a multi-level hierarchy way. CMT (Guo et al., 2022) was composed of a stem architecture to reduce the input size, followed by stacks of convolution layers to extract fine-grained features, and several stacked CMT blocks to model both local and long-range dependencies. BoTNet (Srinivas et al., 2021) proposed a bottleneck transformer by replacing the spatial convolution layer in the ResNet bottleneck block with a global multi-head self-attention module, and three bottleneck transformers took the place of the final three bottleneck blocks in the ResNet backbone. This hybrid design achieved a competitive performance in image classification tasks. CTransCNN (Wu et al., 2023) comprised an initial conv module to extract initial edge and texture features, followed by two parallel transformer-based and CNN-based branches with an information interaction module and a fusion module. Feature maps obtained from the parallel branches are fused in the fusion module to get the multilabel medical image classification result.

2.2. Fine-grained visual classification

The FGVC refers to identifying objects belonging to different sub-categories from the same super-category, such as species of birds, dogs and insect pests. Due to different geographic locations, life cycles and genders, the morphology appearance shows high inter-class similarity and high intra-class dissimilarity, which makes the FGVC task challenging and distinctive. The key point of FGVC is to select the discriminative regions of the target object, and then identify the located discriminative regions instead of the whole input images. Early work (Zhang et al., 2014) leveraged additional bounding boxes and partial annotations to localize the discriminative object parts, which were labor-intensive and time-consuming to annotate such a dataset, therefore limiting their usability and applicability in real-world practice. Consequently, recent works turn their attention to locating informative object parts by using weakly-supervised mechanisms. There are two

major types of weakly supervised-based approaches, namely, the attention-based methods and the localization-based methods.

To better distinguish objects, humans usually selectively concentrate on the salient parts of the objects. Inspired by this, the attention-based methods directly located the discriminative parts by exploiting the attention properties in the output feature maps extracted from the backbone. MA-CNN (Zheng et al., 2017) introduced a channel grouping module to cluster and weight different feature channels with similar response regions into multiple-part attention maps according to the peak response of the feature maps. Moreover, a part classification module was used to learn discriminative fine-grained features from the part-based features. TASN (Zheng et al., 2019) employed a trilinear channel attention module to learn rich fine-grained feature representations. Furthermore, CIN (Gao et al., 2020) designed a self-channel interaction module to learn channel-wise relationships in one image, and a contrastive channel interaction module to explore cross-sample channel-wise relationships in a metric-learning manner to focus on distinctive areas for better FGVC. MMAL-Net (Zhang et al., 2021) proposed two attention modules, called AOLM and APPM, to locate objects and select multiple discriminative parts of images, respectively. All of the aforementioned studies indicated the promising potential of the attention-based methods in FGVC area.

The localization-based methods generally employ a localization subnetwork to find the discriminative areas in images in weakly supervised object detection manners, and then these fine-grained features of the areas are fed into a subsequent classification subnetwork for FGVC task. Ge et al. (2019) combined the Mask R-CNN and CRF-based segmentation methods to get complementary object parts containing rich information. And a bi-directional stacked LSTM was built to fuse and encode these rich complementary features for classification. CDL (Wang et al., 2019) exploited two sub-networks, namely DRG and DFS, to learn the correlation between discriminative regions globally and model the internal spatial correlation in each region locally, respectively. CCFR (Yang et al., 2021) mainly contained two parallel branches. In the top one, a scale-separated NMS module was utilized to extract local discriminative to enhance global features and these regions were then fused to construct a searching database. Meanwhile, another branch calculated the top-n softmax probabilities of the global feature maps and re-ranked the probability results according to the searching database to accomplish classification.

3. Proposed methods

This section will introduce the proposed hybrid CNN-Transformer Architecture for classifying crop pests in large-scale multi-class scenarios. The architecture consists of three components: a hybrid convolution-transformer encoder serving as the basic framework, a dual-path feature aggregation module, and a fine-grained classification module.

3.1. Framework

The Encoder-Decoder structure based on the self-attention mechanism has significant advantages in feature extraction, and many recent studies have applied Transformer to computer vision. ViT splits the input images into fixed-size non-overlapping patches, where the encoder in transformer can capture long-distance dependencies and extract global features of the image by using a self-attention mechanism, significantly improving image classification accuracy. Masked Autoencoders (He et al., 2022) are used to pre-train ViT models by masking the random regions of the input image and then reconstructing the lost RGB regions from visible patches in a self-supervised manner. It is composed of a transformer-based encoder and decoder. The encoder only operates on visible patches, and the decoder processes learnable mask tokens to reconstruct images to learn visual representations. It can learn discriminative and scalable representations from datasets. However, the

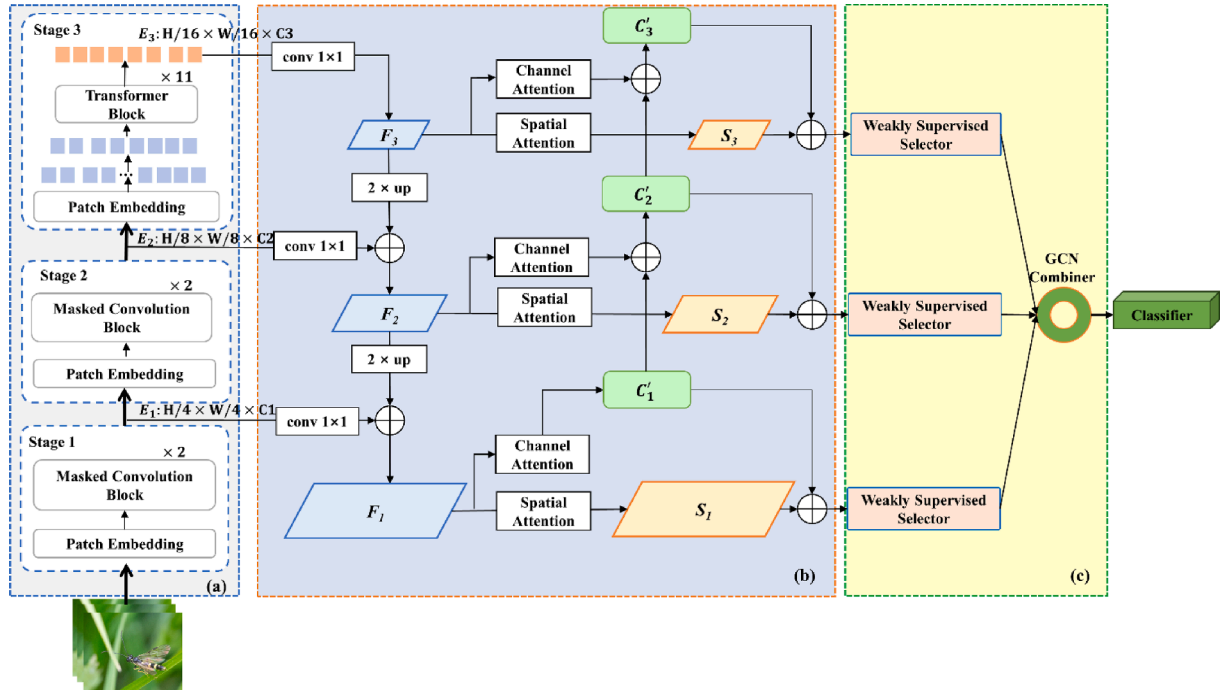


Fig. 1. Architecture of the Pest-Conformer model. (a) Hybrid Convolution-Transformer backbone. (b) Dual-path feature aggregation module. (c) Fine-grained classification module.

native ViT model can only capture the correlation between patches within a single image sample, resulting in insufficient feature extraction ability, which is not conducive to capturing more tiny and multi-scale fine-grained discriminative features, thereby limiting the representation of discriminative feature information. Combining CNN and Transformer, the network can inherit the advantages of CNN and Transformer, reserving the local and global features to the maximum extent. Therefore, by applying MAE to the convolution-transformer network, the ConvMAE (Gao et al., 2022), architecture is able to learn discriminative multi-scale visual representations via the mask auto-encoding scheme.

As shown in Fig. 1, the proposed model comprises a backbone network, a dual-path feature aggregation module, and a fine-grained classification module. By introducing the locality of CNN into the Transformer while reserving its globality, our backbone realizes cascade extraction of visual features from coarse to fine. On the basis of the backbone, a dual-path feature fusion and selection module are further constructed to promote the model to learn more comprehensive, fined, and diverse feature information, thereby enhancing the discriminative representation capabilities of the model. Finally, crop pests are recognized through the Graph Convolutional Network and the last linear classifier.

3.2. Hybrid convolution-transformer backbone

The hybrid convolution-transformer network introduces the inductive bias of CNN to the transformer (Dai et al., 2021), achieving good performance in visual tasks. ConvMAE simplifies these hybrid models. In the early stages, two convolution blocks are used to encode rich local patterns for the high-resolution feature maps. In deeper stages, global self-attention modules in the transformer are adopted to capture high-level semantics and global relationships for low-resolution embedding tokens. Therefore, the encoder with hybrid architecture can learn effective local and global receptive fields at various stages and then generate multi-scale discriminative features.

In our model, as shown in Fig. 1(a), a pre-trained hybrid convolution-transformer encoder architecture is used as the backbone to extract multi-scale features E_1 , E_2 and E_3 , capturing coarse-grained and fine-

grained image information, respectively. The size of output features at different stages are $H/4 \times W/4$, $H/8 \times W/8$, $H/16 \times W/16$, where $H \times W$ is the resolution of the input image. At the beginning of each stage, a Patch Embedding module performs a convolution operation to split the input into patches and linearly project these flattened patches. During the first and second stages, two stacked masked convolution blocks are used to generate Token Embeddings, $E_1 \in \mathbb{R}^{H/4 \times W/4 \times C_1}$ and $E_2 \in \mathbb{R}^{H/8 \times W/8 \times C_2}$, respectively. Each masked convolution block is obtained by replacing the self-attention module with a 5×5 depthwise convolution module. In the third stage, there are 11 global self-attention modules in the transformer block to generate token embeddings, $E_3 \in \mathbb{R}^{H/16 \times W/16 \times C_3}$. The local convolutional layers operations in the first two stages play an essential role in local feature extraction via the convolution operations. The transformer blocks in the third stage use self-attention mechanisms to capture abundant global contextual information and learn long-range dependencies from the coarse-grained features, extending the receptive field from a limited local area to the whole image.

3.3. Dual-path feature aggregation

In the FGVC task, more than relying on the semantic features returned from the backbone network for classification is required. Because it would lose local details in the semantic information of extracted features, which is crucial for improving the recognition performance of FGVC tasks. The Feature Pyramid Network (Lin et al., 2017) is typically used to extract multi-level features. The proposed model selects local detail information in multi-scale features extracted by FPN to generate multi-level fine-grained features. Traditional FPN up-samples deep semantic information and merges with shallow semantic information by element-wise addition to obtain high-resolution, strongly semantic image features. Based on FPN, PANet (Liu et al., 2018) enhances the semantic representation of the backbone by propagating the information in a bottom-up pathway.

Inspired by the FPN, PANet, and AP-CNN network (Ding et al., 2021), this research adopts a Dual-path Feature Aggregation method to extract and fuse multi-scale features, as shown in Fig. 1(b). In the top-down

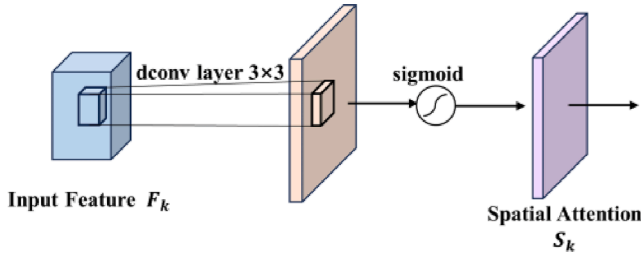


Fig. 2. Spatial attention module.

pathway, an FPN structure is applied to extract three-scale features $\{F_1, F_2, F_3\}$ from backbone on different scales, and transfer high-level feature maps with global context information from higher pyramidal levels to lower pyramidal levels to aggregate low-resolution, semantically robust features with high-resolution, semantically weak features. In a bottom-up way, an attention-based mechanism is used to propagate low-level features with fine details and accurate localization to the entire feature hierarchy. The bottom-up attention pathway is composed of (a) spatial attentions $\{S_1, S_2, S_3\}$, locating discriminative feature areas at different scales, (b) channel attentions $\{C_1, C_2, C_3\}$, calculating channel weights of features and transferring local and spatial information from shallower layers to deep layers through the bottom-up path-way.

Spatial Attention Module. As shown in Fig. 2, a spatial attention map is generated according to the inter-spatial relationships of the input features. It is essential in determining which spatial location to focus on in an image identification task. In the CBAM (Woo et al., 2018) method, both average-pooling and max-pooling operations along the channel axis, respectively, are used to generate two different 2D features at first. Then, these two features are concatenated and convolved by a standard 7×7 convolution operation. In our model, we utilize an improved and simple SAM, directly applying a regular 3×3 deconvolution operation on the input feature F_k to produce a 2D spatial attention map mask. In short, the S_k is calculated as follows:

$$S_k = \sigma(\text{dconv}_{3 \times 3}(F_k)) \quad (1)$$

where σ denotes the sigmoid function, $\text{dconv}_{3 \times 3}$ represents a deconvolution operation with a filter size 3×3 .

Channel Attention Module. As shown in Fig. 3, a channel attention map is generated by exploiting the inter-channel non-linear relationship of the input features through the bottom-up path-way. It can compute channel importance, select critical feature dimensions according to the entire feature maps, and assign weights to each channel. Different from the channel attention mechanism in SE-Net (Hu et al., 2018) and CBAM, we first utilize an adaptive average pool operation to aggregate spatial information and then pass the aggregated information to two fully connected layers to produce the channel attention map mask $C_k \in \mathbb{R}^{C \times 1 \times 1}$, each fully connected layer is followed by a nonlinear function, such as ReLU and Sigmoid. The channel attention mask can be represented as:

$$C_k = \sigma(W_1 \bullet \text{ReLU}(W_2 \bullet \text{AdAvgPool}(F_k))) \quad (2)$$

where σ denotes the sigmoid function, ReLU refers to the ReLU function, and $\bullet$ denotes an element-wise multiplication. W_1 and W_2 are the weight matrices of two fully connected layers.

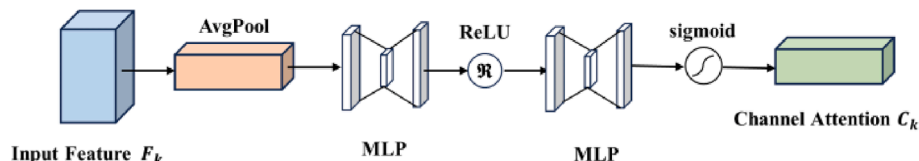


Fig. 3. Channel attention module.

Feature Aggregation. For features in each level of FPN, CAM and SAM modules generate attention feature map masks in channel and spatial dimensions, respectively. Then, the final enhanced feature map is obtained by multiplying the combined feature map masks and the input feature F_k , achieving discriminative features focusing and suppressing background noise information. The enhanced feature map can be represented as follows:

$$f_k = F_k \odot (S_k \otimes C'_k) \quad (3)$$

where $\odot$ denotes an element-wise product and $\otimes$ denotes the broadcasting element-wise addition operation. C'_k is the aggregated channel attention mask through the bottom-up path-way. After the aggregation operation, the low-layer detailed information with more location information is transferred from lower to higher pyramidal levels. The channel aggregation operation can be described as follows:

$$C'_k = \frac{1}{2}(C_k + C_{k-1}) \quad (4)$$

where $k \geq 2$, when $k = 1$, $C'_k = C_k$.

3.4. Fine-Grained classification module

Discriminative Feature Selection. Due to the high inter-class similarity and large intra-class variation of the crop pest classification dataset, high-level features typically have richer contextual information but limited localization capability, resulting in an inability to locate the discriminative areas effectively. Therefore, the key point of FGVC is to learn a local discriminative feature. In attention-based classification methods, attention mechanisms are usually used to locate the discriminative region. Still, various channels of feature maps focus on various pattern information, and some channels might only focus on redundant or background noise information. This study adopts a discriminative region selection strategy based on weakly supervised learning to locate the most discriminative regions while also addressing the issue of redundant information. Inspired by the PIM module (Chou et al., 2022), our main idea is to fuse the high-level and low-level features extracted from the backbone network through a dual path-way in three layers, and a channel and spatial mechanism is used to enhance the entire feature in different scale levels. The enhanced features are then sent to three independent classifiers to calculate classification scores. And then these enhanced feature maps are partitioned into candidate and non-discriminative areas according to the predicted scores. At each pyramidal level, two fully connected layers are used to predict the category of each feature point, following a SoftMax function to calculate the classification probability distribution of predicted results. Then, the prediction probabilities are sorted in descending order, and a certain quantity of feature points with higher probability scores are selected at different levels. The selected feature points are considered to be discriminative feature points that can represent the importance of adjacent regions in the feature map for the FGVC task. Conversely, the unselected feature points are considered to be background or noisy features, which are less beneficial for the FGVC task. Using the discriminative feature points selection method through a weakly supervised selector, our method can pay more attention to the discriminative areas and filter regions with little relevance to crop pest classification. The weakly supervised selection procedure is described as

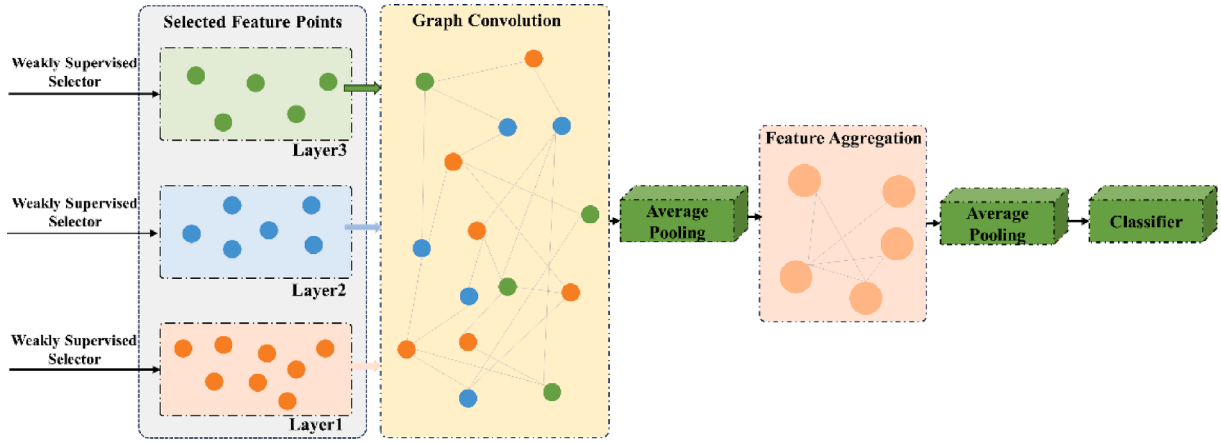


Fig. 4. Procedure of graph convolution and feature aggregation at different pyramid levels.

follows.

The weakly supervised module typically contains a linear classifier composed of two fully connected layers and a SoftMax function. When fed into the selection module, these feature points in f_k are classified at first. Here, $f_k \in \mathbb{R}^{t \times H \times W}$ is used to denote the feature map passed through the liner classifier, where t is the total number of crop pest categories.

$$f_k = W_4(\text{ReLU}(\text{BN}(W_3(f_k)))) \quad (5)$$

where W_3 is the weight matrix of the fully connected layers, W_4 is the weight matrix of the linear classifier, ReLU refers to the ReLU function, BN denotes the batch norm operation.

After calculating the predicted results f_k in the feature map, a SoftMax layer is applied to convert the results into probabilities, which can be considered to be the weakly supervised classification predictions for the crop pest FGVC task.

$$\mathcal{P}(F_k) = \text{softmax}(f_k) \quad (6)$$

where $\mathcal{P}(F_k)$ represents the prediction probability of these feature points in the k th pyramidal level. Since different scale features in different pyramidal levels contribute differently to the final classification result, we choose a certain number of feature points with discriminative features at different feature layers with various scales. In this study, the selection number is set as [128, 64, 32].

Discriminative Feature Aggregation. As shown in Fig. 4, when the discriminative feature points of different scale features in different levels are selected, a Graph Convolutional Network (GCN) is utilized as a feature fusion mechanism to fuse these selected feature points. In the GCN, all these informative feature points can be regarded as the nodes of the graph structure. Therefore, in the graph structure, the node represents various features at different scales and spatial locations, and the edge between the two nodes can be treated as the correlation between two feature points. At first, all selected feature points are fed into the GCN to learn the relationships between different nodes. Then, a pooling layer is adopted to aggregate these features into several super nodes. The selected local features can interact with graph convolution and node aggregation to realize local information flow and integration. Then, they are combined into global discriminative feature points efficiently without corrupting the enhanced feature maps. Finally, an average pooling operation is applied on these super discriminative points, and then these averaged nodes are passed through a linear classifier to accomplish classification.

The main goal of GCN is to learn a nonlinear function g in Graph G . Each GCN layer can be presented as follows:

$$H^{l+1} = g(H^l, A) \quad (7)$$

where $H^0 = X$ denotes the input of the first GCN layer, $X \in \mathbb{R}^{M \times D}$, and M is the total number of nodes, D is the dimension of the feature node. While H^{l+1} represents the node update operation and H^l denotes the current node. A denotes the adjacency matrix.

As mentioned above, all the selected feature points are regarded as graph structure $\mathcal{G}$. The nonlinear function g with layer-wise propagation rule can be expressed as follows:

$$H^{l+1} = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^l W^l) \quad (8)$$

In this GCN model, the node and node itself, the node and its neighbor node, can interact with each other to change local information. Meanwhile, there are implicit stacking convolutional layers in the graph layers, each layer followed by a point-wise non-linearity, so the node is able to fuse information by utilizing multi-layer neighbors. In equation (8), $\tilde{A} = A + I_M$ denotes the adjacency matrix of the graph g with added self-connections, where I_M is an identity matrix, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, W^l is a layer-specific trainable transformation weight matrix. σ represents the sigmoid function. $H \in \mathbb{R}^{N \times M \times t}$, N is the total number of selected feature points at different pyramidal levels, M is the number of the super nodes after aggregation operation, t is the total number of crop pests categories. H^l is the node in the l -th GCN layer, H^{l+1} is the node updating in the $(l+1)$ -th GCN layer.

3.5. Loss function

Feature Extraction Loss. The main goal of our model is to locate a strong discriminative local area to perform crop pest FGVC tasks. It is necessary to make the feature maps extracted from different levels of the backbone have the ability to classify crop pests. At first, the prediction output of all feature points is averaged according to Eq. (9):

$$x_k = \frac{1}{H \times W} \sum_{s \in S} f_{k,s} \quad (9)$$

where $f_{k,s} \in \mathbb{R}^C$ denotes the feature point at position s in the k th pyramidal level of the enhanced features, S represents the output feature space in all levels, and $H \times W$ is the feature size in different levels. The class loss in all pyramidal levels is calculated by utilizing Cross Entropy as follows:

$$L_b = - \sum_{k \in K} \log(x_k^{cls}) \quad (10)$$

Weakly Selection Loss. This loss can be viewed as the discriminative feature selection loss. The location of these selected features in the k th level is regarded as $Mask_k \in \mathbb{R}^{H \times W}$. There are two types of values in $Mask$, where "1" denotes selected areas and "0" represents dropped areas.

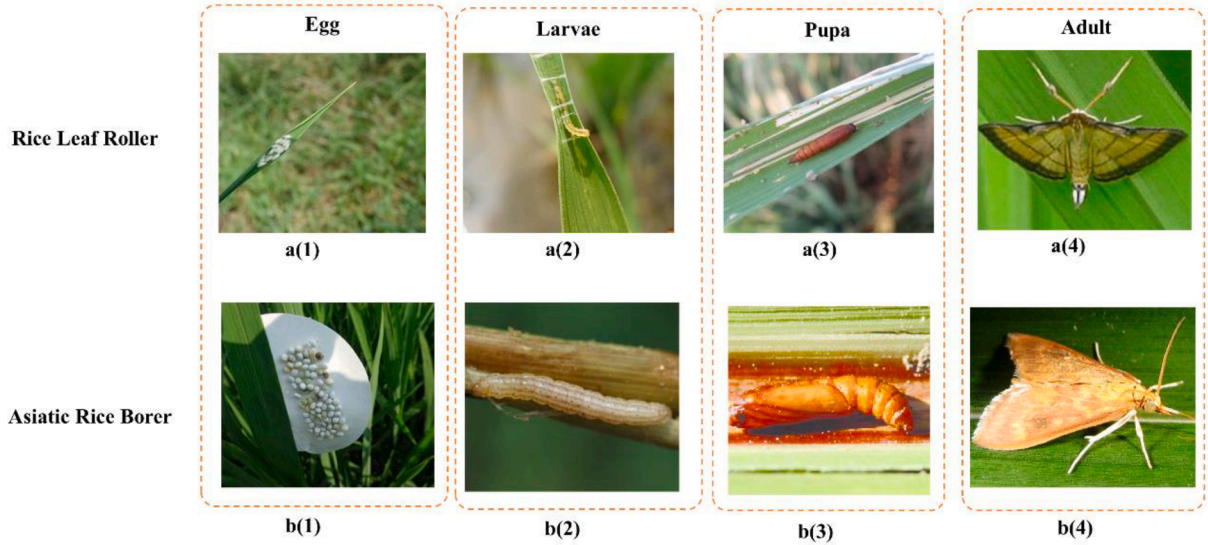


Fig. 5. Sample of IP102 images. Row 1 and 2 show two types of crop pests in different life cycles, respectively. Columns show the morphology of different crop pests in the same life stage.

Therefore, $S \odot (Mask)$ represents the selected points with strong discriminative features, while $S \ominus (Mask)$ denotes the dropped points. Due to the following graph combiner loss L_c has already contained the same function as the weak selection loss, as a result, we use the dropped area selection loss rather than the weak selection loss. The sum of the dropped feature points in the k -th level can be expressed as follows:

$$n_k = \sum_{s \in S \ominus (Mask)} f_{k,s} \quad (11)$$

The dropped area loss function is calculated as follows:

$$L_n = \sum_{k \in K} \sum_{i \in t} \log(1 - n_k^i) \quad (12)$$

where K is the layers of multi-scale feature maps, t is the category number of crop pests in the FGVC task.

Graph Combiner Loss. The discriminative feature points selected by the weakly selection module are fed into the GCN to be fused into multiple super nodes. Subsequently, these super feature nodes are passed through an average pooling layer and sent to a classification head to perform recognition. The graph combiner classification loss is computed by using soft target CrossEntropy. It can be expressed as follows:

$$L_c = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^C y_{it} \log(h_{it}) \quad (13)$$

where N is the total number of nodes in the graph, y_{it} is the probability that node i belongs to each category, and h_{it} is the predicted probability of node i belonging to category t .

In conclusion, the overall loss function is defined as follows:

$$L = \lambda_b L_b + \lambda_n L_n + \lambda_c L_c \quad (14)$$

where λ_b , λ_n and λ_c are the weights of loss functions, respectively. In the training phase, we set $\lambda_b = 1$, $\lambda_n = 0.5$, $\lambda_c = 1$.

4. Experimental results and discussion

4.1. Dataset

We evaluated the proposed method on two datasets. The first dataset used in the study is IP102 (Wu et al., 2019). It's a public dataset of crop pests used for classification and detection tasks. It contains nearly

75,000 images with 102 species of common crop pests, including Alfalfa, Rice, Beet, Citrus, Corn, Mango, Vitis, and Wheat. This dataset was constructed mainly by collecting images from various search engines and websites related to agriculture and insect science. Furthermore, some images containing insect pests were captured from the video clips. According to Wu et al. (2019), the IP102 dataset was divided into three parts: a training set with 45,095 images, a validation set with 7508 images, and a testing set with 22,619 images.

Several practical issues exist in this dataset: (a) images cover the whole pest life cycle, including four stages: eggs, larvae, pupae and adults, which may increase the prediction classification difficulty, especially in the early larvae stages. (b) The dataset exhibits a natural long-tailed data distribution among the number of classes. (c) The dataset has a prominent characteristic of intra-species difference and extensive inter-species similarity. (d) There are many poor-quality images, for instance, visible watermarks and tiny pests on the noisy backgrounds in the dataset. Due to the aforementioned reasons, it is tough to identify the category of each crop pest. Sample images from the dataset are shown in Fig. 5.

The second dataset used in this work is D0 (Xie et al., 2018), which covers nearly 4500 pest images belonging to 40 different pest species captured in field. For D0, we partitioned it into three subsets of training, validation and test. The same splitting manner with (Ayan et al., 2020), 70 % of each specie was randomly chosen for training, the remaining 30 % for validation and 70 % for testing.

1) Data Augmentation

To enhance the diversity of the training data and improve the generalization of the designed network, this study employs an online data augmentation training strategy. This approach aims to address the issues of overfitting and enhance the suitability of the model for identifying crop pests in natural environments. Each image within a batch undergoes a sequence of random data augmentation combination procedures to obtain a training set.

In the training phase, the original images are pre-processed in the following steps:

Step 1: First, all images are cropped into a random scale size with a random aspect ratio of the original image and then resized to 224×224 pixels.

Step 2: All images are randomly and horizontally flipped with a given probability.

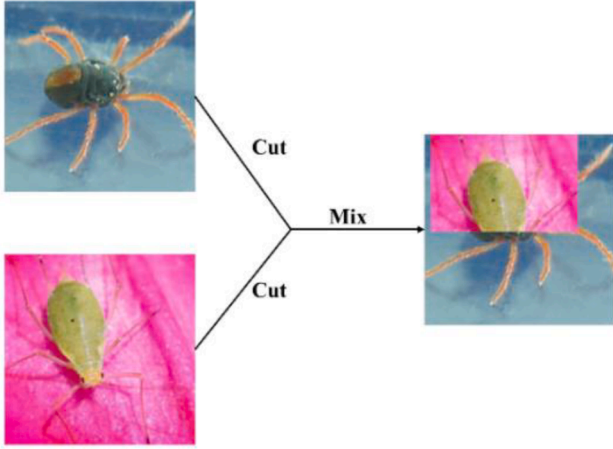


Fig. 6. Schematic of CutMix augmentation.

Step 3: All images are applied AutoAugment (Cubuk et al., 2019) augmentation policy with settings of “rand-m9-mstd0.5-inc1”.

Step 4: The brightness, contrast, saturation, and hue of the images’ color are randomly changed by using the color jitter transform;

Step 5: A rectangle region is randomly selected and erased in an image using random erasing data augmentation.

Step 6: The CutMix (Yun et al., 2019) augmentation strategy is applied to the mini-batch in an iteration.

But in the validation phase, only resize and center crop are used.

2) CutMix Augmentation

The CutMix augmentation method is a regularization strategy to improve the performance of classification and localization. It performs transformation by removing a local area of one training image and replacing it with a same-size patch from another image in the same batch to produce a new locally natural image, as shown in Fig. 6. Meanwhile, the ground-truth label of the image is generated by proportionally mixing two labels according to the area ratio of the two combined parts.

In each iteration of the training phase, a combined new sample $(\tilde{x}, \tilde{y})$ is produced by cutting and mixing two randomly chosen samples in one batch as follows:

$$\begin{aligned}\tilde{x} &= M \odot x_A + (1 - M) \odot x_B \\ \tilde{y} &= \lambda y_A + (1 - \lambda) y_B\end{aligned}\quad (15)$$

where $M \in \{0, 1\}^{W \times H}$ indicates the rectangular binary mask regions

where to be cut and pasted from two selected images, and $\odot$ denotes element-wise product. λ is the combination ratio sampled from the uniform distribution (0, 1). The aspect ratio of the binary rectangular mask M is proportionate to the original image, and the bounding box coordinates of M are sampled from the uniform distribution according to the following formula:

$$\begin{aligned}r_x &\sim \text{Unif}(0, W), r_w = W\sqrt{1 - \lambda} \\ r_y &\sim \text{Unif}(0, H), r_h = H\sqrt{1 - \lambda}\end{aligned}\quad (16)$$

where (r_x, r_y, r_w, r_h) is the bounding box coordinates.

4.2. Evaluation metrics

Due to the imbalanced distribution of pest categories in the IP102 dataset, the Accuracy (Acc) and weighted F1 Score, in terms of evaluation metrics, are chosen to evaluate the classification performance in this work.

For the crop pests FGVC model, accuracy is the most important performance metric. The ratio of correctly predicted pest categories in samples is known as accuracy. It is calculated as follows:

$$\text{Acc} = (TP + TN) / (TP + TN + FP + FN) \times 1 \quad (17)$$

Precision is used for validating the predicted results. It measures how precise the model is. The precision represents the proportion of predicted pest categories that are correctly labeled. It is calculated as follows:

$$\text{Pre} = TP / (TP + FP) \times 100\% \quad (18)$$

Recall measures the model’s ability to predict actual positive pest classes. The recall reveals the probability of the correctly predicted pest classes. It is calculated as follows:

$$\text{Re} = TP / (TP + TN) \times 100\% \quad (19)$$

F1-score is a weighted reconciliation and averaging of Precision and Recall. It is usually used to evaluate model classification performance comprehensively. Moreover, the weighted F1 score is calculated by taking the weighted average of the class-wise F1 scores, while the weight denotes the number of samples available in that class. It is calculated as follows:

$$\text{F1} = 2 \bullet \text{Pre} \bullet \text{Re} / (\text{Pre} + \text{Re}) \times 10 \quad (20)$$

$$\text{F1}^* = (\sum_{i=1}^L \text{F1}_i \bullet \omega_i) / L \quad (21)$$

In the above formulas, TP , FP , TN and FN represent the number of

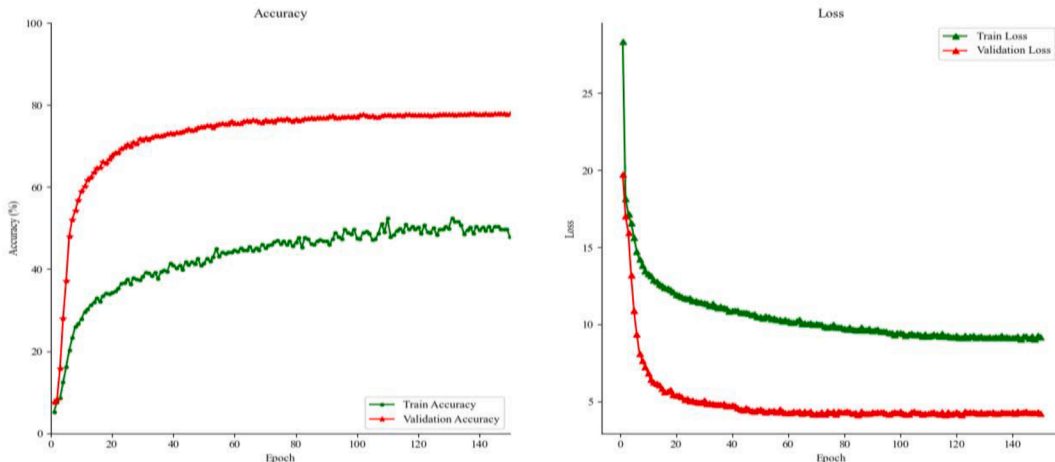


Fig. 7. Experiment results of the proposed method on IP102.

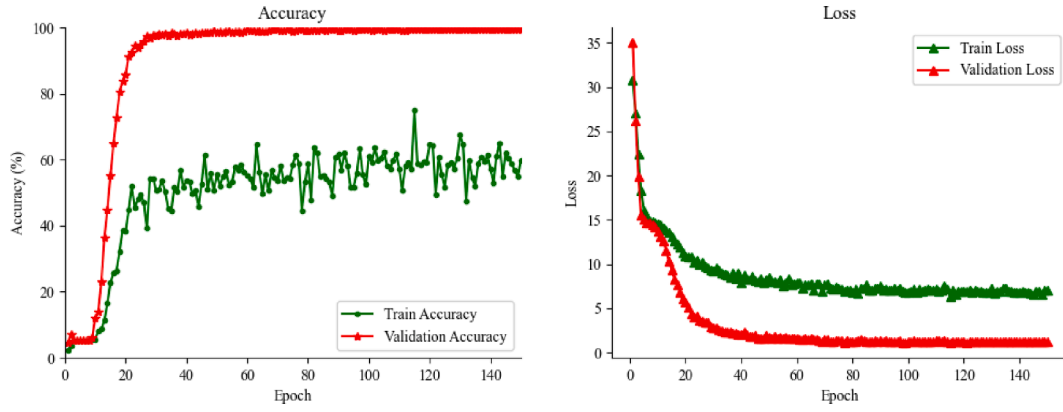


Fig. 8. Experiment results of the proposed method on D0.

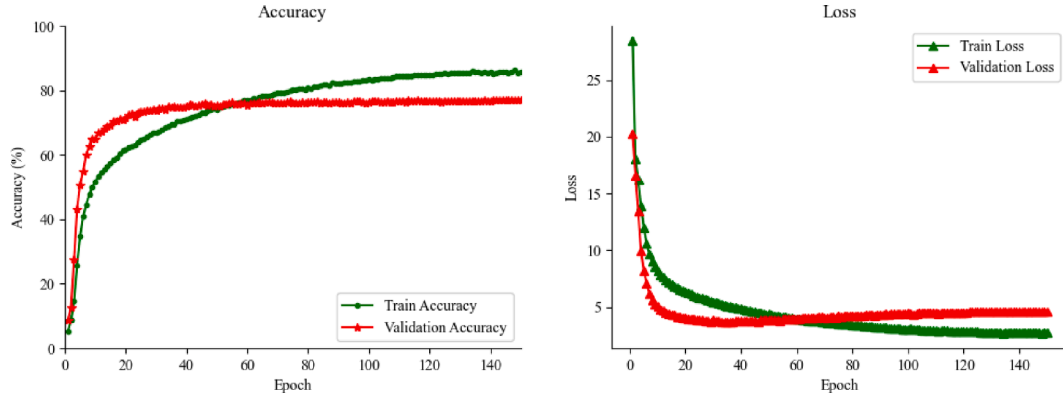


Fig. 9. Experiment results of the proposed method without CutMix on IP102.

True Positive, False Positive, True Negative and False Negative, respectively. Where i denote the category index, L represents the total number of categories, and ω_i is the weight value, which refers to the ratio of samples in the dataset belonging to the i -th class.

4.3. Experimental setup and training

All training and validation experiments are conducted on a deep learning workstation performed on an Ubuntu 20.04 LTS system with an Intel Xeon Gold 6130 CPU@2.1 GHz $\times$ 16 processors, two CUDA devices with dual Nvidia GeForce RTX 3090 GPUs, and 32 GB memory. The software is implemented on the PyTorch 1.13.0 framework and the PyTorch Image Models library with Python 3.9.

To speed up the training time of our model, the AdamW optimizer is adopted with a learning rate lr using the liner scaling rule: $lr = \text{base_lr} \times \text{batchsize}/256$, a base lr of 0.0005, a weight decay of 0.05, a batch size of 32, and the layer-wise learning rate decay is set as 0.65. Adam optimizer is considered because it incorporates weight decay and L2 Regularization; it can effectively prevent overfitting problems during the training phase. Regarding the learning rate schedule strategy, a cosine annealing with a warmup algorithm is applied to iteratively update the network parameters based on a batch of training examples, the warmup epoch is set to 5, and the total number of epochs is set as 150.

Transfer learning is applied in our training procedures, the experimental results can be seen in Fig. 7 and Fig. 8 on IP102 and D0 datasets, respectively. The two graphs showed that the proposed method was a little underfitting, the same as (Setiawan et al., 2022). In the accuracy graph, the accuracy performance results in validation data were better than the training data. However, it could be seen that the use of CutMix augmentation was the main reason of underfitting from Fig. 9. As explained above, a combined new training samples was generated by

cutting and mixing two randomly selected samples in one training batch, which makes the generated image and label look complicated. Especially in the crop pest dataset, the size of some insect pests is much smaller than the surrounding backgrounds (seen in Fig. 5), once the CutMix augmentation outperforms, it's difficult to choose the proper area ratio of the two images. As a result, the proposed model will experience confusion during training phrase, and the training accuracy shows smaller than validation accuracy.

4.4. Compared with state-of-the-art methods

Based on the literature survey, the methods used in pest recognition on the IP102 dataset are divided into three categories: CNN-based, Ensemble-based, and Transformer-based. To validate the classification effectiveness of the proposed model, its performance is compared to the SOTA (State-of-the-art) methods. Table 1 shows the comparison results and the calculation consumption, where the best recognition results are in boldface. The results show that our method achieves 77.81 % accuracy and 77.36 % F1-score on the dataset. However, due to the hybrid complicated CNN-transformer architecture, the proposed method had a large number of parameters and consumed the highest floating-point operations in crop pest recognition, which means the proposed method has the highest model complexity and feature extraction capability.

Compared to the CNN-based models, it is a remarkable fact that our model outperforms these methods by a wide margin. The accuracies of these CNN models, such as the ResNet-50 (Wu et al., 2019), and its variants, i.e., FR-ResNet (Ren et al., 2019), STN-ResNet (Yang et al., 2021), AM-ResNet (Zhang et al., 2022) and DMF-ResNet (Liu et al., 2020), EfficientNet V2-S (Zheng et al., 2023), MobileNetV2 (Setiawan et al., 2022) and its variants, PCNet (Zheng et al., 2023), IRNV2 (Ayan,

Table 1
Comparison of the best performance on the IP102 dataset.

Category	Model	Accuracy	F1 Score	Params (M)	FLOPs (G)
CNN-Based Models	ResNet-50 (Wu et al., 2019)	0.4950	0.4010	25.64	4.09
	FR-ResNet (Ren et al., 2019)	0.5524	0.5418	30.78	–
	STN-ResNet (Yang et al., 2021)	0.7329	–	–	5.45
	MobileNetV2 (Setiawan et al., 2022)	0.7132	–	3.50	0.30
	AM-ResNet (Zhang et al., 2022)	0.7299	–	–	–
	DMF-ResNet (Liu et al., 2020)	0.5922	0.5837	29.70	–
	EfficientNet V2-S (Zheng et al., 2023)	0.7000	–	20.31	2.80
	PCNet (Zheng et al., 2023)	0.7370	–	20.70	2.88
	IRNV2 (Ayan, 2024)	0.7184	0.6406	60.40	–
	VRFNet (Nandhini & Brindha, 2024)	0.6834	0.6834	–	–
Transformer-Based Models	FRFC+LSMAE (Liu et al., 2022)	0.7469	0.7436	85.80	17.60
	CNN+Transformer (Peng & Wang, 2022)	0.7489	–	28.31	–
	DWViT-ES (Hechen et al., 2024)	0.7600	–	19.60	3.5
Ensemble Models	GAEnsemble (Ayan et al., 2020)	0.6713	0.6576	–	–
	Ensembles with Six CNN models (Nanni et al., 2022)	0.7411	0.7290	–	–
	Ours	0.7781	0.7736	87.37	25.61

Table 2
Comparison of model performance on D0 dataset.

Category	Model	Accuracy	F1 Score	Params (M)	FLOPs (G)
CNN-Based Models	MobileNet-Small (Nandhini & Brindha, 2024)	0.9777	0.973	2.54	0.06
	DenseNet121 (Nandhini & Brindha, 2024)	0.9589	0.9589	2.83	7.98
	EfficientNet-B0 (Nandhini & Brindha, 2024)	0.9833	0.9833	5.29	0.39
	Deep CNN (Thenmozhi & Reddy, 2019)	0.9597	–	–	–
	MNV2 (Ayan, 2024)	0.9989	0.9986	3.50	0.30
	VRFNet (Nandhini & Brindha, 2024)	0.9912	0.9912	–	–
	Ours	0.9951	0.9950	87.37	25.61
Transformer-Based Models	CNN + Transformer (Peng & Wang, 2022)	0.9915	–	28.30	–
Ensemble Models	GAEnsemble (Ayan et al., 2020)	0.9881	0.9881	–	–
	Ensembles with Six CNN models (Nanni et al., 2022)	0.9981	0.9971	–	–
	Ours	0.9951	0.9950	87.37	25.61

2024), VRFNet (Nandhini & Brindha, 2024), are less than 74 % on IP102 dataset. The dataset creator (Wu et al., 2019) first tested a fine-tuning pre-trained ResNet50 and reached the highest accuracy rate of 49.50 %. Following their work, some variants of ResNet50 were designed to overcome the complexity of this dataset. FR-ResNet, modifying the original ResNet architecture layer, obtained an accuracy of 55.24 %. Taking advantage of the attention mechanism in focusing on useful features and suppressing unnecessary ones, STN-ResNet, a ResNet equipped with a spatial transformer combined with channel and spatial attention mechanisms, obtained an insect pest classification accuracy of 73.29 %. DMF-ResNet, a deep multibranch fusion ResNet obtained by stacking the multi-branch fusion residual block to learn the multi-scale representation, achieved an accuracy of 59.22 %. From the perspective of network energy consumption, AM-ResNet, a low-energy-consumption Addition-Multiplication hybrid ResNet structure, obtained an accuracy of 72.99 % while reducing 40 % of energy consumption. Some researchers turned their attention to lightweight CNN models, a modified MobileNetV2, with modern and sophisticated training tricks, such as dynamic learning rate, CutMix augmentation, freezing layers, and sparse regularization, which obtained a competitive accuracy of 71.32 %. Recently, PCNet, an EfficientNet-V2 variant incorporating a coordinate attention mechanism to learn inter-channel and positional information, obtained the highest recognition accuracy of 73.7 %, outperforming the other CNN-based models. In the same work, the original EfficientNet-V2 was also tested on the IP102 dataset, which obtained an accuracy of 70 %. IRNV2, abbreviation for InceptionResNetV2, combined the advantages of ResNet and Inception. By using a genetic algorithm-based optimization strategy to determine transfer learning and fine-tuning hyperparameters, IRNV2 acquired an accuracy of 71.84 %. VRFNet utilized patch-based augmentation and multiscale feature fusion to extract rich semantic details in insect pest images, it achieved an accuracy of 68.34 % on IP102. In short, by taking advantage of vision transformers in modelling global interactions, our model is nearly 4 percent superior to the SOTA CNN-based methods.

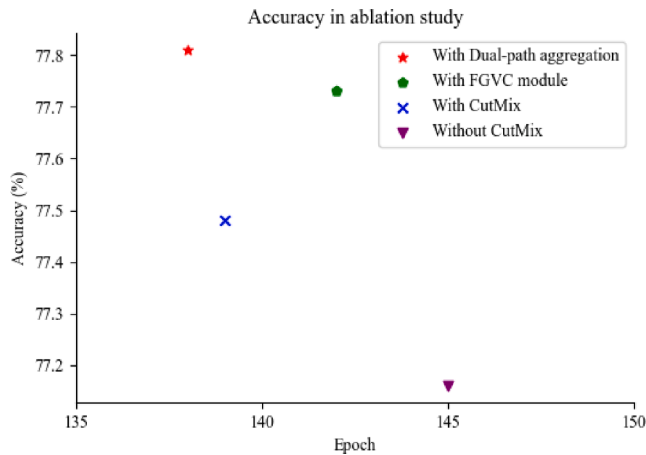
Another way is to combine the predictions of several CNNs, taking advantage of the strengths of each CNN. These ensemble methods typically perform better than individual CNN. GAEnsemble (Ayan et al., 2020), combining three best-performed models by weighted voting, achieved an accuracy of 67.13 %. The best ensemble reported by Nanni et al., (2022) obtained a SOTA performance. This ensemble method combined six CNN models using different Adam variants algorithms for optimization based on DGrad. It achieved an accuracy of 74.11 %. Ensemble methods may achieve better accuracies based on sophisticated training techniques; they may have shortcomings, inefficient feature representation, and may be both computing resources and time-consuming.

Besides selecting the classical CNNs as the backbone to build a dedicated crop pest recognition model, some models built upon vision transformer architecture were also proposed. Liu et al., (2022) designed a transformer auto-encoder based on the latent semantic masking model for pest classification. The selected ViT-B/16 model was pre-trained on ImageNet based on LSMAE. Then the fine-tuned transformer-based encoder was utilized to extract insect features, and a multilayer perceptron head was used for classification. The accuracy obtained in this study was 74.69 %. A transformer is effective in capturing and modeling rich global information; therefore, it can improve the performance of insect classification accuracy. Peng and Wang (2022) designed a transformer-based network by exploring CNN as the backbone and an attention-based classification head. Their hybrid CNN and transformer model achieved a classification accuracy of 74.89 %. Focusing on the computation complexity of Transformers, DWViT-ES (Hechen et al., 2024) introduced an efficient-self-attention mechanism to reduce model parameters and computational cost, it achieved a competitive accuracy of 76.00 % with nearly 20 % reductions of computation cost. However, by exploiting fine-grained features and aggregating multi-scale features, our method achieves nearly 2 % improvement compared with the

Table 3

Results of ablation experiments.

Model	Factors				Accuracy	F1 Score
	Without CutMix	With CutMix	With FGVC module	With Dual-path aggregation		
Pest-Conformer	✓				0.7716	0.7671
		✓			0.7748	0.7696
		✓	✓		0.7773	0.7732
		✓	✓	✓	0.7781	0.7736

**Fig. 10.** Accuracy in ablation study.

classification accuracies of these transformer-based models.

As seen in Table 1, it is clearly shown that fine-tuning a transformer-based network outperforms traditional CNNs based on the IP102 dataset. Compared with CNNs, the transformer-based models are typically featured with larger model size. However, the main challenge occurs in the fact that this model strongly depends on extensive datasets for training on a big scale. By leveraging the hybrid CNN Transformer architecture and self-supervised pre-training on the extensive ImageNet-1 K dataset, our model is able to accurately identify distinctive features and accurately represent the spatial structure and fine-grained details. An alternative method involves employing Ensemble learning, which combines many deep learning-based models to create a more powerful model that surpasses them. This is accomplished by utilizing contemporary sophisticated training techniques to produce competitive or superior performance in pest categorization. Nevertheless, this approach has encountered challenges in terms of computing complexity, inadequate time consumption, and ineffective feature representation.

In order to better estimate the performance of the proposed model, we also evaluated the model on a relatively small dataset D0. As shown in Table 2, existing models have achieved excellent recognition accuracy in D0 Dataset. Especially the MNV2 (Ayan, 2024) method, a lightweight MobileNetV2 model together with genetic algorithm-based hyper-parameters optimization, obtained the highest accuracy on D0 dataset. Meanwhile, the VRFNet (Nandhini & Brindha, 2024), CNN-Transformer hybrid model (Peng & Wang, 2022) and ensemble model (Nanni et al., 2022) also achieved excellent accuracies over 99 %. However, our proposed method acquired competitive accuracy of 99.52 %, which was superior to the CNN-Transformer hybrid model, and was slightly lower than the MNV2 and the ensemble model. We think this phenomenon may be due to the limited number of training samples, only nearly 3500 images, which is not enough to train a large model with beyond 80 million parameters. As a result, lightweight model with fewer parameters will perform better on small dataset, such as D0.

Compare Table 1 with Table 2, we found that the performance of all models in the D0 dataset is superior to IP102 dataset. That's because the images in D0 dataset are high quality, and well balanced. More

importantly, the number of pest spices is only 40, which is far less than the number of pest spices in IP102.

4.5. Ablation study

Several ablation studies are performed to demonstrate the effectiveness and necessity of the novel tricks and new designs in our proposed methods. Table 3 and Fig. 10 shows the ablation experimental results of Pest-Conformer on the IP102 dataset. In the first experiment, a pre-trained multi-scale hybrid CNN-Transformer encoder is chosen as the backbone to fine-tune on the dataset to validate the classification performance. It achieves accuracy of 77.16 % and F1-score of 76.71 %. It's shown that the performance has already shown noticeable improvement compared to previous studies. The main reason for this is that combining the transformer with convolution effectively brings locality into vision transformers, which are good at capturing long-range dependencies but lack the ability to extract local patterns in visual tasks. As a result, convolution blocks in high-level are applied to encode local information at an early stage, and the transformer blocks in low-level are used to realize global information aggregation at a late stage. Therefore, the proposed hybrid backbone can learn more discriminative multi-scale features by aggregating local and global feature maps at different stages, enhancing discriminative feature extraction ability. Another reason is beneficial to the ability of masked autoencoder architecture that reconstructs the original masked image patches from partial observation of the image patches. Our method can learn discriminative and scalable feature representations.

In the following experiment, based on the above experiment, a CutMix augmentation strategy is carried out in the training phase. The results indicate that CutMix helps increase recognition accuracy to 77.48 %. On the one hand, CutMix helps make the backbone focus on discriminative features all over the target regions, and it can guide the classifier to make a classification based on a wider series of features by learning spatially distributed representations. On the other hand, a larger number of tiny pests exist in the dataset where the complicated background environment dominates the image. As a consequence, two patches that only contain a background will be combined to generate a new one in some extreme situations. This will make CutMix work ineffectively. However, in this experiment, the results show that our model is improved by using CutMix augmentation.

The third experiment adds an FGVC module to our hybrid backbone. The same as the former two experiments, the FGVC module also makes improvements to our model. The accuracy is 77.73 %, and the F1 score is 77.32 %. This indicates such a module can effectively select proper regions from different scales of pixel-level feature maps and then filter out feature points with strong discrimination. These discriminative feature points are fused with a graph combiner to obtain prediction results. It means that the fused discriminative features can realize performance improvements of the proposed pest FGVC model.

In the last experiments, a dual-path feature aggregation module adhered to the model based on the third experiment. The proposed method achieves the best classification performance on the IP102 dataset, and the accuracy is 77.81 %. For such a result, the following reasons need to be considered. For one thing, an FPN is chosen as the base top-down structure to extract different scale features. The lower pyramidal level obtains higher resolution features by upsampling

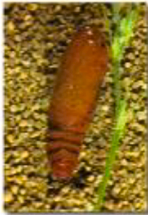
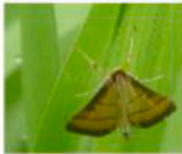
Egg			
	Confidence 0.692	0.255	0.515
	Prediction Rice leaf roller	Yellow cutworm	Yellow rice borer
Ground Truth	Rice leaf roller	Black cutworm	Asiatic rice borer
Larva			
	Confidence 0.679	0.757	0.628
	Prediction Rice leaf roller	Asiatic rice borer	Black cutworm
Ground Truth	Rice leaf roller	Asiatic rice borer	Large cutworm
Pupa			
	Confidence 0.615	0.755	0.398
	Prediction Rice leaf roller	Asiatic rice borer	Large cutworm
Ground Truth	Rice leaf roller	Asiatic rice borer	Yellow cutworm
Adult			
	Confidence 0.694	0.682	0.787
	Prediction Rice leaf roller	Asiatic rice borer	Yellow cutworm
Ground Truth	Rice leaf roller	Asiatic rice borer	Yellow cutworm

Fig. 11. False recognition samples in IP102.

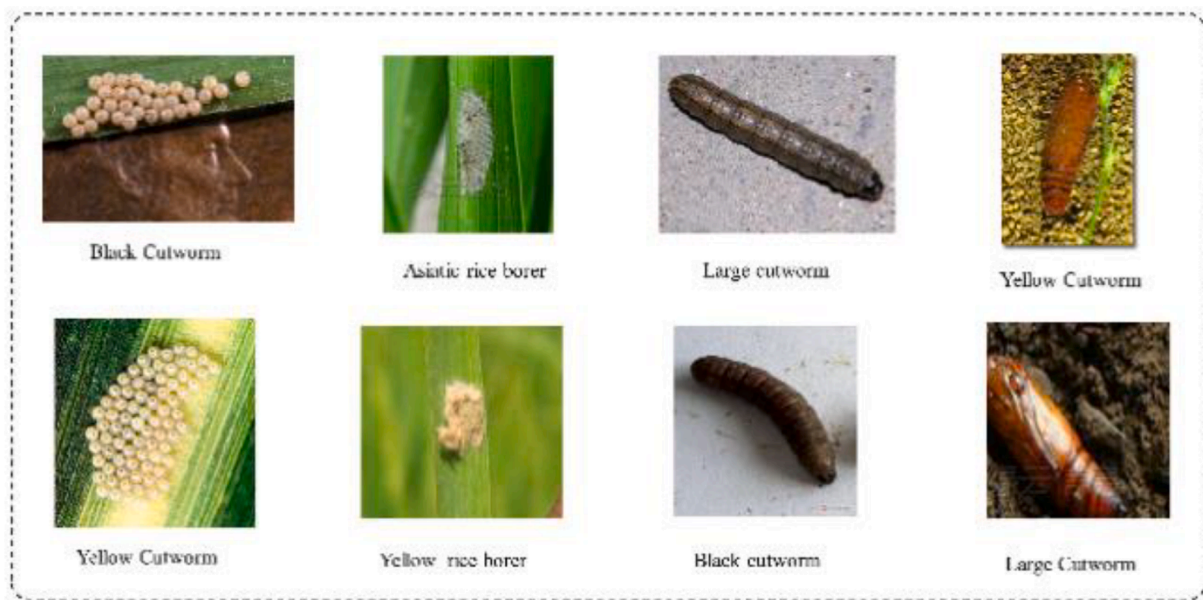


Fig. 12. Morphological similarity of false recognition samples.

spatially coarser but semantically more robust feature maps from the higher pyramidal level, which benefits the classification task. For another, a bottom-up attention structure is introduced based on the pyramidal features; the spatial attention modules are used to locate discriminative feature areas from different scales, and the channel attention modules are used to exploit inter-channel relationships, capture channel correlations, and transfer more local details in a bottom-up manner. The dual-path and multi-scale attention mechanism makes the low-level features propagate to the high-level effectively to enrich the entire feature hierarchy, enhancing the fine-grained feature representation of the crop pest image and increasing the localization ability of local discriminative features.

4.6. Qualitative analysis

As shown in Fig. 5, due to the ecological characteristics, the same category of crop pest appears different in different life cycle, and different subcategories may have similar forms in the same growth stage in IP102 dataset. According to Hechen et al. (2024), transformer-based model performed worse in the egg stage, followed by the larval stage. The recognition accuracies in pupa and adult stage are little difference, which are relatively higher than the other two phases. There are several reasons for this: (1) in the egg and larval stage, the insect pest size is much smaller than the background size, and it's very difficult to identify such tiny and dense objects; (2) in the pupa and adult stage, the size of insect pest increased, and some crop pests had obvious color and texture features, this will improve the performance accuracy. However, due to the feature of inter-class similarity, the accuracy improvement was limited.

In order to better evaluate the fine-grained feature discrimination capability of the proposed model, several false recognition samples are shown in Fig. 11. It can be seen that Pest-ConFormer fail to effectively recognize crop pest in the growth stages of egg, larva and pupa. However, in the adult stage, our method performs well. The main reasons are as follows: (1) the imbalanced image number in different life cycle. As compared to adult stage, the image number of the other three growth stages is relatively small, our model fails to learn discriminative features in the images; (2) the characteristic of inter-class similarity. Different crop pests in the same growth stage have the same appearance, and this will make the model confused. Fig. 12 shows the morphological similarity of false recognition samples in Fig. 11. Obviously, the crop pests in

the same column are look like similar. Therefore, in order to improve the classification accuracy in early growth stage, more early-stage sample images are needed in the training phrase.

5. Conclusion and future work

Vision-based recognition of large-scale crop pests, especially in the larval stage, is one of the important and effective means for early-stage pest monitoring, which also is one of the most important ways for crop protection. As a mainstream crop pest recognition technology, the conventional deep learning approaches have faced various challenges, including the abundance of crop pests, the presence of complicated background interference, the high similarity between different pest classes, and the significant diversity in the appearance of pests within the same class. In order to address these issues, this work introduces a novel CNN-transformer based fine-grained crop pest recognition network called Pest-ConFormer. Pest-ConFormer utilizes a hybrid architecture that incorporates a dual feature aggregation pathway. This pathway consists of an FPN-like top-down structure and a PANet-like bottom-up attention structure. The purpose of this pathway is to transfer both high-level global context information and low-level local detailed information. This transfer enriches the features at each level. A fine-grained classification module is implemented, comprising a weakly discriminative feature selection module and a feature aggregation module based on GCN. This module aims to select discriminative feature points and combine global discriminative features to further improve the performance of FGVC. The recognition performance of the proposed method is assessed using the IP102 benchmark dataset, and several ablation experiments are conducted to validate the effectiveness of each module. The experimental results show that the proposed method yields an accuracy of 77.81 % and an F1-score of 77.36 %, nearly 2 percent points superior to the SOTA crop pest recognition method.

Although the proposed model in this paper can achieve the highest accuracy in terms of fine-grained classification on IP102, it still has shortcomings in high computation cost. In future work, our plan involves creating a lightweight transformer module and implementing the lightweight model on mobile devices to facilitate early-stage monitoring in the field.

CRediT authorship contribution statement

Mingwei Fang: Conceptualization, Methodology, Writing – original draft. **Zhiping Tan:** Conceptualization, Data curation, Writing – original draft. **Yu Tang:** Supervision, Writing – review & editing. **Weizhao Chen:** Validation. **Huasheng Huang:** Investigation. **Sathian Dananjayan:** Formal analysis. **Yong He:** Writing – review & editing. **Shaoming Luo:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code will be available at <https://github.com/mwfang/pestconformer>.

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